

Week 5 Make a rule work · paper route

Paper route · GROW Computing · 40 minutes

Name: _____ Date: _____

Week 5 paper route. Make a rule work.

Goal: I can make Player return to the start when it touches Maze.

This is the whole of Week 5. It is not a rehearsal. It is not a warm-up for a computer day.

You do every part of Week 5 here, on paper.

You do not need a computer. You do not need to save a file.

You need this pack, a pen, and one counter.

The counter is your Player.

The grey bands on the big grid are Maze.

Your teacher watches you work. Your teacher writes down what you did.

You can point, say, draw or write. An adult can write your exact words for you.

Take a pause when you need one. Come back to the same line.

Point, say, draw or write. Someone may record your exact words.

On the screen route this week is saved as W05_Walls.sb3. On the paper route no file is made. Your pages are the record: write your name and the date on every page and put them in your folder. An adult initials the folder. Nobody should ask you for W05_Walls.sb3 if you took the paper route.

Write your code

Player · THE NEW WEEK 5 SCRIPT. The wall rule. Player keeps checking. If Player touches Maze, Player goes back to the start. Build this one yourself. Do not copy a finished script.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- forever
- if then
- touching [Maze]?
- go to x [start x] y [start y]
- change x by 10
- change y by 10

Player · Keep this script from Week 4. It moves Player right. Write it again here so your whole sprite is on paper.

Write one block on each line. Start with the hat block, which is done for you.

when right arrow key pressed

Blocks you may use this week:

- change x by 10
- change x by -10
- change y by 10
- change y by -10

Player · Keep this script from Week 4. It moves Player left.

Write one block on each line. Start with the hat block, which is done for you.

when left arrow key pressed

Blocks you may use this week:

- change x by 10
- change x by -10
- change y by 10
- change y by -10

Player · Keep this script from Week 4. It moves Player up.

Write one block on each line. Start with the hat block, which is done for you.

when up arrow key pressed

Blocks you may use this week:

- change x by 10
- change x by -10
- change y by 10
- change y by -10

Player · Keep this script from Week 4. It moves Player down.

Write one block on each line. Start with the hat block, which is done for you.

when down arrow key pressed

Blocks you may use this week:

- change x by 10
- change x by -10
- change y by 10
- change y by -10

Player · Keep this start script. It puts Player at the start. It is a second green flag hat. It is not the same script as the wall rule.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- go to x -180 y -120
- go to x [start x] y [start y]
- change x by 10

Player (the broken example - do NOT keep this one) · THE BROKEN EXAMPLE from the Starter. Write it out so you can see the fault. This script checks one time only. There is no forever. You study it, then you draw a ring round the missing part. You do not keep this script.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- if then
- touching [Maze]?
- go to x -180 y -120
- forever

The maze

The big printed grid is the Scratch stage at its real size. x goes from -240 on the left to 240 on the right. y goes from -180 at the bottom to 180 at the top. Every square is 10 across, because change x by 10 moves Player 10.

The grey bands are Maze. They are printed at their real places:

- bottom wall: x -220 to 220, y -160 to -140
- top wall: x -220 to 220, y 140 to 160
- left wall: x -220 to -200, y -140 to 140
- right wall: x 200 to 220, y -140 to 140
- left inside wall: x -70 to -50, y -140 to 80
- right inside wall: x 50 to 70, y -80 to 140

The yellow square at x 168 to 192, y 108 to 132 is Finish. You do not use Finish this week.

Your counter is Player. Put the dot in the middle of the counter on the crossing point you are told.

THE TOUCHING RULE. Say it out loud before you start:

If any part of your counter is on grey, that is touching Maze.

Even a small part counts.

If no part of your counter is on grey, that is not touching Maze.

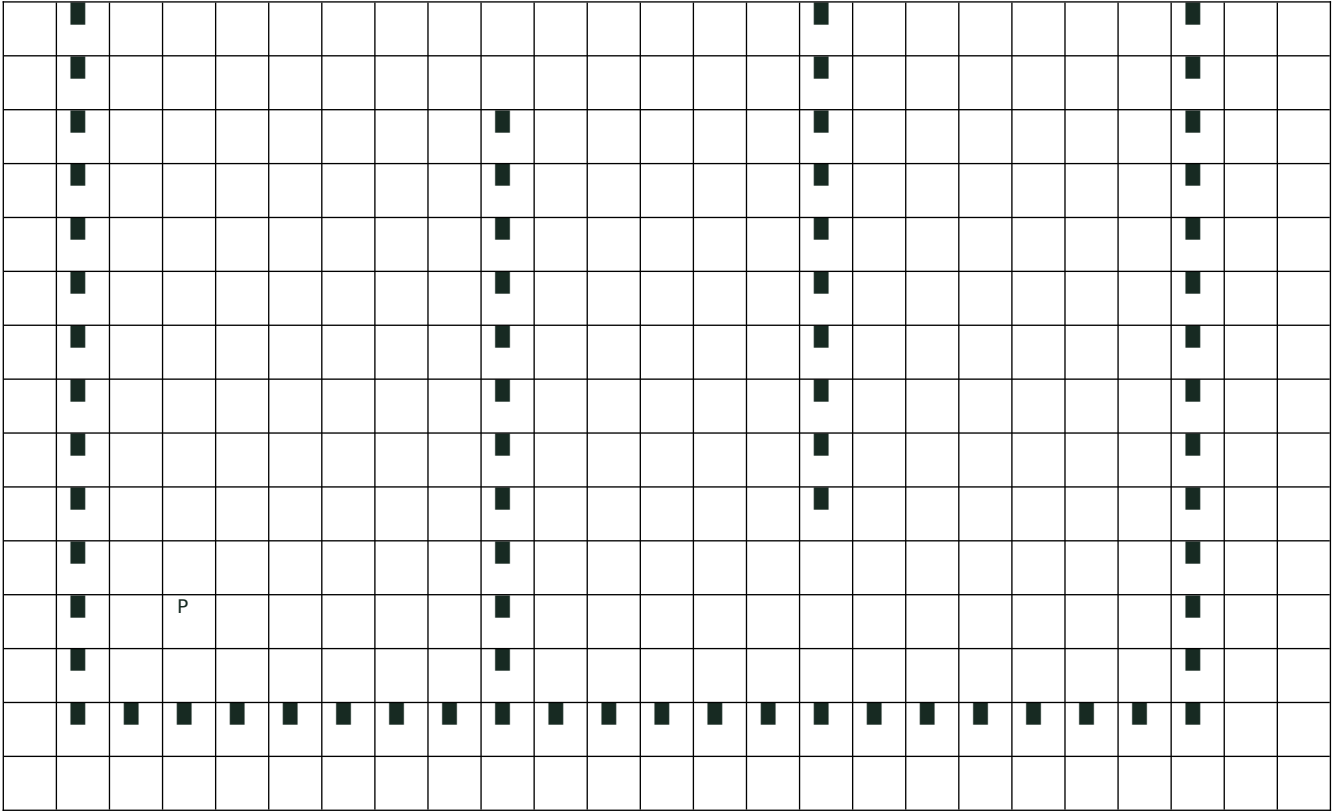
WHAT YOU DO WITH THE COUNTER THIS WEEK:

1. Put the counter on the start: x -180, y -120. Check the start is clear of grey.
2. Place-and-check. Put the counter on x 0, y 0. Say Yes or No. Then x -60, y 0. Then x 60, y 0. These are the same three test places the screen lesson uses.
3. Walk the clear space. From the start, move the counter up one square at a time, three times: x -180 y -110, then x -180 y -100, then x -180 y -90. Check after every move.
4. Walk into wall one. Put the counter back on the start. Move left one square: x -190, y -120. Check. Move left again: x -200, y -120. Check. This one is touching. Now do what your script says: slide the counter back to x -180, y -120.
5. Walk into wall two. Place the counter on x 40, y 0. Check. Move right one square: x 50, y 0. Check. This one is touching. Slide the counter back to x -180, y -120.
6. Corner test (Challenge only). Put the counter on x -200, y -140. That is where two walls meet.

Every time you move the counter, you must check again. That is what forever means.

The maze below is the real one, at its real size. One square is 20 steps. Player starts at (-180, -120). Finish is at (180, 120). Move a counter square by square and write down where it lands.

[illegible]



Predict, trace, compare

Predict first. Put your counter on each place. Say or write Yes or No BEFORE you look carefully. Then look and fill the next box.

Put the counter here	My prediction: touching Maze? Yes or No	What the grid shows: touching Maze? Yes or No	What my script does next
x 0, y 0 (the open middle)			
x -60, y 0 (the left inside wall)			
x 60, y 0 (the right inside wall)			
x -180, y -120 (my start)			

Trace 1. A clear walk. Start at x -180, y -120. Press up three times. Row 1 is done for you. Fill in every empty box.

Step	Block I run	x after	y after	Touching Maze? Yes or No	Where the counter goes
1	when green flag clicked, then go to x -180 y -120	-180	-120	No	Counter on x -180, y -120
2	forever, then if touching Maze?				
3	up arrow, then change y by 10				
4	if touching Maze?				
5	up arrow, then change y by 10				
6	if touching Maze?				
7	up arrow, then				

	change y by 10				
8	if touching Maze?				

Trace 2. Wall one. Start at x -180, y -120. Press left twice. Row 1 is done for you.

Step	Block I run	x after	y after	Touching Maze? Yes or No	Where the counter goes
1	when green flag clicked, then go to x -180 y -120	-180	-120	No	Counter on x -180, y -120
2	forever, then if touching Maze?				
3	left arrow, then change x by -10				
4	if touching Maze?				
5	left arrow, then change x by -10				
6	if touching Maze?				
7	go to x [start x] y [start y] - this block only runs if step 6 was Yes				
8	if touching Maze? (the loop checks again)				

Trace 3. Wall two. This is a different wall. Place the counter on x 40, y 0. Press right once. Row 1 is done for you.

Step	Block I run	x after	y after	Touching Maze? Yes	Where the counter
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				or No	goes
1	Place the counter at the test place	40	0	No	Counter on x 40, y 0
2	forever, then if touching Maze?				
3	right arrow, then change x by 10				
4	if touching Maze?				
5	go to x [start x] y [start y] - this block only runs if step 4 was Yes				
6	if touching Maze? (the loop checks again)				

Compare. One check, or a check that repeats. Use the SAME counter walk for both: start at x -180, y -120, then left, then left again.

Question	The broken example (no forever)	My script (with forever)
At the green flag, the counter is on x -180, y -120. Touching Maze?		
The counter moves to x -190, y -120. Does the script check here?		
The counter moves to x -200, y -120. Does the script check here?		
Where does the counter finish?		
Which script does the job I want?		

Challenge. Only after your wall rule is finished. Change the return place to a bad one. Predict the fault. Trace it.

What I changed	My prediction	What happened on the grid
I changed the return place to x -60, y 0. That place is inside a wall.		
I put the counter on x 50, y 0. Touching Maze? Yes. Where does the counter go now?		
The forever loop checks again straight away. Touching Maze? Where does the counter go?		
I repaired the return place. My clear start is x _____, y _____.		
I retested the corner at x -200, y -140.		
I retested an ordinary wall at x 50, y 0.		

My safe start. Fill this in and use the same numbers in every go to block on your sheets.

My safe start	x	y	Is any part of the counter on grey?	Is there a clear square on every side?
The start I chose				
The model start (you may use this one)	-180	-120		

My completion record

What this week shows

AQA 71638 outcome 3: Observe the pupil constructing the collision script, then demonstrate contact at two wall positions. Record the actual consequence and any prompts.

- ☐ I built a repeated touching Maze check on Player. I wrote it in my Player grid: when green flag clicked, then forever, then if touching Maze? then go to my start.
- ☐ The consequence runs only when the condition is true. I showed this on my trace sheets. On my No rows the go to block did not run. On my Yes rows it did.
- ☐ I tested two wall positions and a clear space. Wall one was x -200, y -120. Wall two was x 50, y 0. My clear space was x 0, y 0 and my clear walk up from the start.
- ☐ I saved and reopened the working rule. I wrote my name and the date on my sheets. I put them in my folder. Later I found my own sheets on my own. I traced one wall test again from my own writing, and it still worked.
- ☐ I said what I think about the return place. An adult wrote what I said. I know what changed because of it.

Adult record

- ☐ I watched this pupil write the collision script block by block in the Player grid. I did not give them a finished script to copy. I did not use the teacher model as something to copy from.
- ☐ I watched the pupil put if then inside forever, and put the return block inside if then. The nesting is shown by the indents, or by the boxes the pupil drew, or by the pupil pointing at the two spaces and telling me which block goes in each.
- ☐ I watched the pupil name Maze in the sensing block, not Finish.
- ☐ I watched the pupil demonstrate contact at TWO DIFFERENT wall positions with the counter. Position one: x -200, y -120 (the left wall). Position two: x 50, y 0 (the right inside wall). These are two different walls, not the same wall twice.
- ☐ THE ACTUAL CONSEQUENCE I OBSERVED: at each contact the pupil moved the counter back to their own start, x ____ , y ____ . I recorded what actually happened, not what should have happened.
- ☐ I watched the pupil check a clear space and show me that the return did NOT run there.
- ☐ PROMPTS: I recorded, against each pupil action, whether it was unprompted, or given a verbal prompt, or given a modelled step. I did not leave the prompt box blank.
- ☐ I recorded the pupil's access method: writing / speaking / pointing / drawing / directing an adult to move the counter / exact words scribed. Where the pupil directed an adult, the decision was the pupil's and the adult only carried it out.
- ☐ I recorded whether the pupil went back to the check row on their own, or whether I told them to.
- ☐ ROUTE TAKEN: PAPER. This pupil completed Week 5 on paper. That is recorded here, in this Teacher Guide, and nowhere else.

- ☐ I have not written on, altered or reprinted the AQA Summary sheet in any way.
- ☐ I completed the AQA Summary sheet for outcome 3 from my own individual observation of this pupil. I did not infer creation from supplied code.

Adult name: _____ Date: _____

Route taken: paper · Recorded in the teacher guide, not on the AQA form.